

HUNDRED DAYS OF CODING - 6

❖ QUESTION 1)

Write a C program to find the radius, diameter, and circumference of a circle when its area is given.

We must create a C program that will take area as an input and print radius, diameter and circumference as output. The formulas we will need to print output are as follows:

$$\text{Radius} = r = \frac{\sqrt{\text{area}}}{\pi}, \quad \text{Diameter} = \text{radius} * 2, \quad \text{Circumference} = 2 * \pi * \text{radius}$$

• INPUT / OUTPUT TABLE

Test Case	Input(Area)	Expected Output	Actual Output	Result
1	28	Radius = 2.99 Diameter= 5.97 Circumference = 18.75	Radius = 2.99 Diameter= 5.97 Circumference = 18.75	Pass
2	54	Radius = 4.15 Diameter = 8.29 Circumference= 26.04	Radius = 4.15 Diameter = 8.29 Circumference= 26.04	Pass
3	100	Radius = 5.64 Diameter= 11.29 Circumference= 35.44	Radius = 5.64 Diameter= 11.29 Circumference= 35.44	Pass

- **ALGORITHM**

- 1) Start
- 2) Declare variables area, radius, diameter and circumference.
- 3) Input the area of the circle.
- 4) Calculate the radius using the following formula:
$$r = \frac{\sqrt{area}}{\pi},$$
- 5) Calculate the diameter using the following formula:
$$D = 2 * radius$$
- 6) Calculate the circumference using the following formula:
$$Circumference = 2 * \pi * radius$$
- 7) Display the radius, diameter, circumference.
- 8) Stop

- **C Program**

```
#include<stdio.h>
#include<math.h>
#define PI 3.14
int main()
{
    float radius, diameter, area, circumference;
    printf("Enter the area of the circle");
    scanf("%f", &area);

    radius = sqrt(area/PI);
    diameter = 2*radius;
    circumference = 2*PI*radius;

    printf("\nRadius = %.2f", radius);
    printf("\nDiameter = %.2f", diameter);
    printf("\nCircumference = %.2f", circumference);

    return 0;
}
```

Compile it using:

```
gcc hdoc6.c -o hdoc6 -lm
```

Run it using:

```
./hdoc6
```

- **Screenshots/Snippets of compilation and execution:**

```
(kali㉿kali)-[~/Desktop/HDOC]
$ vi hdoc6.c

(kali㉿kali)-[~/Desktop/HDOC]
$ gcc hdoc6.c -o hdoc6 -lm

(kali㉿kali)-[~/Desktop/HDOC]
$ ./hdoc6
Enter the area of circle28

Radius = 2.99
Diameter = 5.97
Cirumference = 18.75

(kali㉿kali)-[~/Desktop/HDOC]
$ ./hdoc6
Enter the area of circle54

Radius = 4.15
Diameter = 8.29
Cirumference = 26.04

(kali㉿kali)-[~/Desktop/HDOC]
$ ./hdoc6
Enter the area of circle100

Radius = 5.64
Diameter = 11.29
Cirumference = 35.44

(kali㉿kali)-[~/Desktop/HDOC]
$
```

❖ QUESTION 2)

Write a C program to find the inner radius, outer circumference, and area of an annulus when its area and outer radius are given.

We must create a C program that will take area and radius of an annulus as input and print inner radius, outer circumference and area as output using the following formulas:

Given: Area of Annulus = A, Outer Radius = R

Inner Radius(r) : $r = \sqrt{R^2 - A/\pi}$

Outer Circumference = $2\pi R$

Inner Circumference = $2\pi r$

• INPUT / OUTPUT TABLE:

Test Case	Input	Expected Output	Actual Output	Result
1	Area(A) = 104 Outer Radius = 10	Inner Radius = 8.18 Outer Circumference = 62.80 Inner Circumference = 51.36	Inner Radius = 8.18 Outer Circumference = 62.80 Inner Circumference = 51.36	Pass
2	Area(A) = 156 Outer Radius = 12	Inner Radius = 9.71 Outer Circumference = 75.36 Inner Circumference = 60.99	Inner Radius = 9.71 Outer Circumference = 75.36 Inner Circumference = 60.99	Pass
3	Area(A) = 204 Outer Radius = 20	Inner Radius = 18.30 Outer Circumference = 125.60 Inner Circumference = 114.95	Inner Radius = 18.30 Outer Circumference = 125.60 Inner Circumference = 114.95	Pass

• ALGORITHM:

- 1) Start.
- 2) Input the area of the annulus and the outer radius.
- 3) Calculate the inner radius.
- 4) Calculate the outer Circumference.
- 5) Calculate the inner Circumference.
- 6) Display the calculated values.
- 7) Stop.

• C program:

```
#include <stdio.h>
#define PI 3.14
#include <math.h>
int main()
```

```
{
```

```
float area, outerRadius;
float innerRadius;
float outerCircumference;
float innerCircumference;

printf("Enter area of annulus: ");
scanf("%f", &area);
printf("Enter outer radius: ");
scanf("%f", &outerRadius);

innerRadius = sqrt((outerRadius * outerRadius) - (area / PI));
outerCircumference = 2 * PI * outerRadius;
innerCircumference = 2 * PI * innerRadius; printf("\nInner Radius = %.2f",
innerRadius);
printf("\nOuter Circumference = %.2f", outerCircumference);
printf("\nInner Circumference = %.2f", innerCircumference);

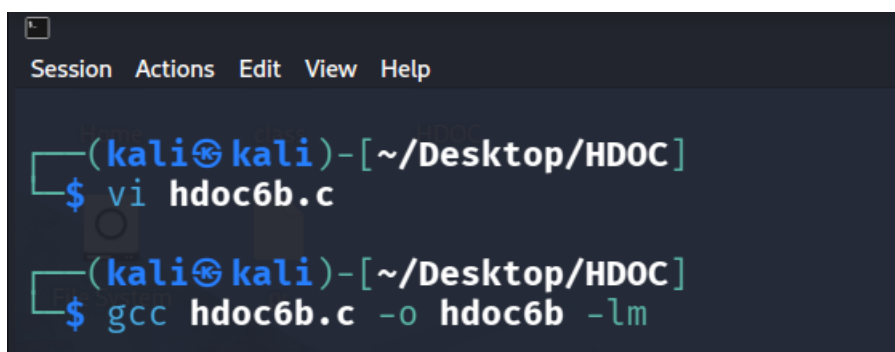
return 0;
}
```

Compile it with:

```
gcc hdoc6b.c -o hdoc6b -lm
```

Run it using:

```
./hdoc6b
```



```
Session Actions Edit View Help

(kali㉿kali)-[~/Desktop/HDOC]
$ vi hdoc6b.c

(kali㉿kali)-[~/Desktop/HDOC]
$ gcc hdoc6b.c -o hdoc6b -lm
```

OUTPUTS:

```
(kali㉿kali)-[~/Desktop/HDOC]  
$ gcc hdoc6b.c -o hdoc6b -lm
```

```
(kali㉿kali)-[~/Desktop/HDOC]  
$ ./hdoc6b  
enter the area of the annulus104  
enter the outer radius10
```

```
Inner Radius = 8.18  
Outer Circumference = 62.80  
Inner Circumference = 51.36
```

```
(kali㉿kali)-[~/Desktop/HDOC]  
$ ./hdoc6b  
enter the area of the annulus156  
enter the outer radius12
```

```
Inner Radius = 9.71  
Outer Circumference = 75.36  
Inner Circumference = 60.99
```

```
(kali㉿kali)-[~/Desktop/HDOC]  
$ ./hdoc6b  
enter the area of the annulus204  
enter the outer radius20
```

```
Inner Radius = 18.30  
Outer Circumference = 125.60  
Inner Circumference = 114.95
```